



SAVEETHA
SCHOOL OF ENGINEERING
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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SMART EMERGENCY RESPONSE AND AMBULANCE ROUTING SYSTEM

CSA0311– Data Structures

Guided By:
Dr. Kruthika

Presented By)
Madhumitha V(192521332)
Sweta R(192511408)

INTRODUCTION

- Enables quick and efficient ambulance dispatch during medical emergencies to save lives during critical situations.
- Uses data structures to manage emergency requests and ambulance records.
- Finds the shortest route using graph-based Dijkstra's Algorithm.
- Improves emergency response time through optimized routing and resource management.



PROBLEM STATEMENT

- Manual ambulance dispatch can delay emergency response during critical situations.
- Lack of an optimized route increases travel time during medical emergencies.
- Managing ambulance availability without an organized system is inefficient and time-consuming.
- Delayed medical assistance can reduce the chances of saving patients' lives.



OBJECTIVES



- ❖ Develop a centralized system to manage emergency requests efficiently.
- ❖ Assign the nearest available ambulance to reduce response time.
- ❖ Find the shortest route using graph-based pathfinding algorithms.
- ❖ Improve emergency services through optimized data management and routing.



LITERATURE REVIEW



S.No	Author & Year	Method Used	Findings	Limitation
1	Kumar et al. (2024)	Graph-Based Routing	Improved ambulance route optimization	Does not consider live traffic
2	Sharma et al. (2024)	Dijkstra's Algorithm	Reduced travel distance and response time	Performance decreases on large road networks
3	Lee et al. (2024)	Queue-Based Dispatch System	Efficient emergency request processing	Queue delays during peak emergencies
4	Ahmed et al. (2024)	Hash Table Management	Faster ambulance record retrieval	Requires frequent data updates
5	Patel et al. (2025)	Tkinter-Based Emergency System	Simple and user-friendly interface	Limited scalability for large cities

PROPOSED SOLUTION



- Build a digital emergency response system for efficiently handling emergency requests.
- Maintain ambulance records using efficient data structures for quick access.
- Implement Dijkstra's Algorithm to calculate the shortest and fastest travel route.
- Provide a simple Tkinter-based graphical interface for easy user interaction.



Module 1 - Emergency Request & Ambulance Management



Goal

- To efficiently receive emergency requests, maintain ambulance records, and dispatch the nearest available ambulance.

Key findings

- Register and manage emergency requests received from patients.
- Store ambulance details and track their availability efficiently.
- Process emergency requests using a queue for systematic handling.
- Search ambulance records using hashing and assign the nearest available ambulance.



Module 2 – Route Optimization & Navigation

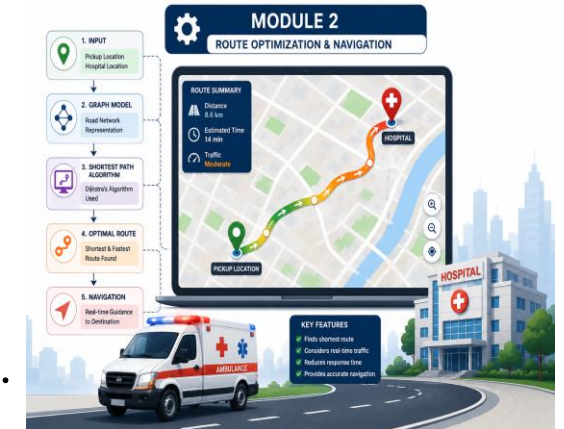


Goal:

To determine the shortest and fastest route for the assigned ambulance to reach the emergency location efficiently.

Key findings:

- Represent the road network using a graph data structure for efficient mapping.
- Calculate the shortest path using Dijkstra's Algorithm for faster navigation.
- Provide optimized navigation from the ambulance to the destination in real time.
- Reduce travel time and improve overall emergency response efficiency.



SDG Goals



1. SDG 3 – Good Health and Well-being

- Enables faster emergency medical response.
- Reduces ambulance response time.
- Improves patient safety during emergencies..

2. SDG 9 – Industry, Innovation and Infrastructure

- Uses efficient data structures for emergency management.
- Optimizes ambulance routing with Dijkstra's Algorithm.
- Supports smart healthcare infrastructure.

3. SDG 10 – Reduced Inequalities

- Improves emergency transportation services.
- Enhances hospital–ambulance coordination.
- Supports smarter and safer healthcare systems.



OUTPUT





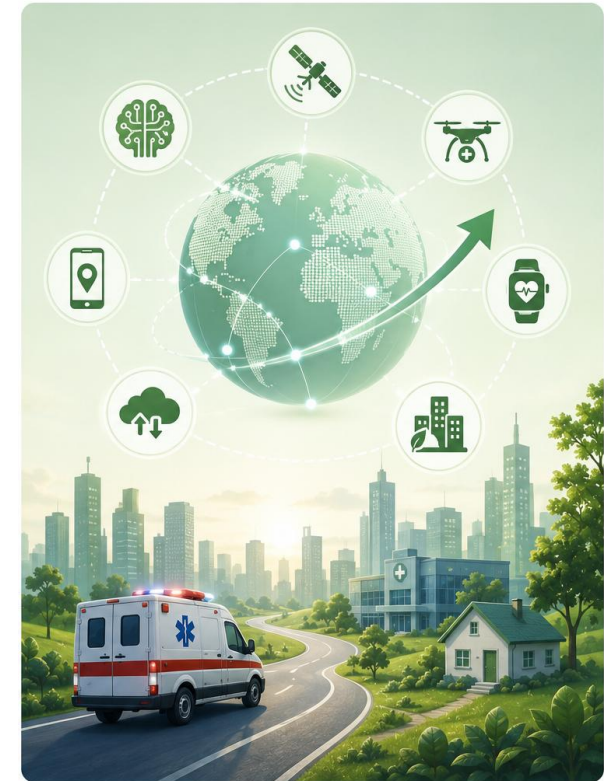
OUTPUT



ADVANTAGES



- Reduces emergency response time through faster ambulance dispatch.
- Finds the shortest and most efficient route to the destination.
- Improves ambulance management using efficient data structures.
- Provides a simple, user-friendly interface for easy operation.



CONCLUSION

- Provides a fast and efficient ambulance dispatch management system.
- Optimizes route planning using graphs and Dijkstra's Algorithm.
- Improves emergency response through efficient data structure implementation.
- Enhances patient safety by reducing ambulance response and travel time.



FUTURE SCOPE



- Integrate GPS technology for real-time ambulance tracking .
- Add live traffic updates for smarter route planning.
- Develop a mobile application for patients and ambulance staff.
- Expand the system to support multiple hospitals and emergency service networks.



REFERENCE



1. Shen, Y., Li, D., Wang, W., & Dong, X. (2024) Blood Glucose Concentration Prediction Based on Double Decomposition and Deep Extreme Learning Machine Optimized by Nonlinear Marine Predator Algorithm. *Mathematics*, 12(23), 3708. ([MDPI][1])
2. Nemat, H., Khadem, H., Elliott, J., et al. (2024) Data-driven blood glucose level prediction in type 1 diabetes: a comprehensive comparative analysis. *Scientific Reports*, 14, 21863. ([Nature][2])
3. Allam, F., et al. (2024) Using nonlinear auto-regressive with exogenous input neural network (NNARX) in blood glucose prediction. *Bioelectronic Medicine*, 10(11). ([Springer][3])
4. Han, Y., Kim, D., Woo, J., & Kim, J. (2024) Glu-Ensemble: An ensemble deep learning framework for blood glucose forecasting in type 2 diabetes patients. *Heliyon*, 10(8), e29030. ([PubMed][4])
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6. Moon, K., Kim, J., Yoo, S., et al. (2025) Personalized blood glucose prediction in type 1 diabetes using meta-learning with BiLSTM-transformer hybrid model. *Scientific Reports*, 15. ([Nature][7])



Engineer to Excel



THANK
YOU

Data structure (1)

Home Insert Play Design Transitions Animation

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Mevalurkuppam, Tamil Nadu, India 

Ece Department, Mevalurkuppam, Tamil Nadu 602105, India

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